

CHAINLAB

Crypto Wallets and Key Management

Custody models, signing and practical protection

FREE BLOCKCHAIN EDUCATION RESOURCE

This guide is designed for learning and revision. It explains foundational concepts and highlights practical limitations and safety considerations.

Learning objectives

- Compare custodial and self-custody models.
- Explain seed phrases, private keys and transaction signing.
- Build a practical wallet security routine.

1. Custody models

- With custodial services, a provider controls key material or transaction authority on behalf of users. This can simplify access but introduces provider, account and legal risks.
- With self-custody, the user controls keys and bears responsibility for backup, device security and transaction approval.

2. Seed phrases and keys

- A seed phrase can derive wallet keys. Anyone who obtains it may be able to control associated assets; never enter it into an unsolicited website or share it with another person.
- Keep backups offline in secure locations. Consider physical threats, loss, damage and inheritance planning.

3. Transaction review

- Confirm network, destination, amount, fees and any approvals or permissions. Read signing prompts carefully and stop if the request is unexpected or unclear.
- A transaction may be irreversible. Sending to the wrong address or approving malicious permissions can result in permanent loss.

4. Operational security

- Use device updates, screen locks, reputable wallet software and unique passwords for related accounts.
- Separate high-value storage from experimental activity where appropriate. Be cautious with browser extensions, clipboard substitution and fake support messages.
- Use a small test transaction when appropriate, but remember that a successful test does not guarantee every later transaction is safe.

5. Recovery planning

- Document recovery steps without exposing secrets. Consider how trusted beneficiaries could locate instructions while keeping key material protected.
- Test backups carefully and understand whether wallet recovery depends on a passphrase or other additional factor.

Review questions

- Who is responsible for a lost self-custody seed phrase?

- What details should be checked before signing?
- Name four common wallet attack methods.

Educational notice: This material is for general education only. It is not financial, legal, tax or investment advice. Verify current network and project details before acting.